



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

Conventional environmental sustainability platforms are limited by historical logging, manual data entry, and generic recommendations, leaving a critical research gap in active, personalized decision support and multi-resource optimization. To address these limitations, this project proposes EcoSphere, an intelligent, machine learning-driven environmental decision support platform designed to transition sustainability management from passive tracking to proactive optimization. The architecture employs a React.js client interface communicating with a Spring Boot middleware gateway for secure PostgreSQL and PostGIS spatial database management, paired with a Python FastAPI microservice to handle the machine learning workload. The analytical layer utilizes XGBoost and scikit-learn regression models to forecast future carbon, water, and electricity consumption profiles based on historical user behavior. Conflicting resource objectives are resolved using a Google OR-Tools constraint solver to optimize tradeoffs between carbon footprint, utility consumption, and financial costs. A household digital twin enables comparative simulations of alternative sustainability strategies, while a generative artificial intelligence connector provides natural-language explanations of decisions. Additionally, an evidence verification framework audits user activities utilizing geocoding, timestamp verification, and image-based validation. By integrating behavioral forecasting, multi-objective optimization, and transparent AI reasoning, EcoSphere enables individuals and institutions to achieve measurable, verifiable resource reductions.

KEYWORDS: Environmental Decision Support, Machine Learning, Multi-Resource Optimization, Behavioral Forecasting, Digital Twin Simulation, Explainable Artificial Intelligence, Evidence Verification, Sustainability Intelligence